SUBRATA MALLICK

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PROFESSIONAL SUMMARY

ECE graduate with hands-on experience building full-stack and AI-powered applications using Java, Spring Boot, MySQL, Python, React/Next.js and REST APIs. Experienced in backend architecture, multimodal AI integration, machine learning, database design and production deployment. Seeking an entry-level software development role.

TECHNICAL SKILLS

Languages: Java, C++, Python, JavaScript, TypeScript, C
Backend: Spring Boot, Spring Data JPA, Hibernate, REST APIs, FastAPI, DTOs, Validation, Exception Handling
Frontend: HTML, CSS, JavaScript, React, Next.js, Tailwind CSS, Recharts
Database: MySQL, SQL
AI/ML: scikit-learn, Machine Learning, Face Recognition, InsightFace, Gemini Multimodal API, Prompt Engineering
Tools: Git, GitHub, Gradle, Postman, Swagger/OpenAPI, IntelliJ IDEA
Core: DSA, OOP, DBMS

EDUCATION

B.Tech — Electronics and Communication Engineering | Academy of Technology | 2022–2026 | CGPA: 7.95
Higher Secondary (12th) | Bhatpara Amarkrishna Pathsala (H.S.) | 2022 | 80.8%
Secondary (10th) | Bhatpara Amarkrishna Pathsala | 2020 | 88.85%

PROJECTS

SchoolBuddy.ai — AI-Powered Multimodal Educational Tutor | Java, Spring Boot, React/Next.js, MySQL, Gemini API
Live: https://schoolbuddy-frontend.vercel.app
GitHub Backend: https://github.com/MallickSayan407/schoolbuddy-backend
GitHub Frontend: https://github.com/MallickSayan407/schoolbuddy-frontend

- Developed a full-stack AI tutoring platform for Grade 1–10 students supporting text and image-based questions with grade-aware responses across Math, Science and English.
- Built a Spring Boot REST backend with MySQL conversation persistence, DTO-based APIs, validation, centralized exception handling and contextual multi-turn conversations.
- Integrated multimodal Gemini AI with a six-model fallback cascade, grade-specific prompting and six learning modes (Explain, Solve, Hint, Simplify, Practice, Quiz), with math response normalization for LaTeX formatting and step-by-step solutions.
- Developed a responsive Next.js/React frontend with image upload, Markdown/KaTeX rendering and real-time AI interaction; deployed on Vercel, Render and Aiven MySQL and regression-tested across Grades 1, 5 and 10 and all six learning modes.

Cricket Analytics | Java 21, Spring Boot, MySQL, Python, React, JavaScript, Recharts
Live: https://cricket-analytics-two.vercel.app
GitHub: https://github.com/MallickSayan407/CricketAnalytics

- Built a full-stack cricket analytics platform covering ODI, Test, T20I and IPL data, with more than 11,000 matches, 8,400 players and 600 venues, plus player comparison, leaderboards, filtering and visualization.
- Integrated a FastAPI/scikit-learn ODI win-probability model (51 features; 61.7% accuracy and 67.4% ROC-AUC on a 488-match locked test set) and a Monte Carlo tournament simulator.
- Deployed as a multi-service application: React on Vercel, Spring Boot API, FastAPI ML service and MySQL on Railway.

Face Attendance System | Java 21, Spring Boot, MySQL, Python, InsightFace, HTML, CSS, JavaScript
GitHub: https://github.com/MallickSayan407/face-attendance-system

- Developed a full-stack employee attendance system using layered architecture, REST APIs, image upload, duplicate prevention and daily summary reporting (present/absent counts and attendance percentage).
- Integrated Python InsightFace (512-dimensional face embeddings, 0.6 similarity threshold) with Spring Boot, marking attendance only after 5 recognition confirmations; implemented DTOs and centralized exception handling.

Employee Management System | Java, Spring Boot, MySQL, HTML, CSS, JavaScript
GitHub: https://github.com/MallickSayan407/employee-management

- Implemented 10 REST endpoints covering CRUD, name/department/salary search and multi-criteria filtering, with DTOs, validation of 7 fields, centralized exception handling, pagination, sorting and Swagger/OpenAPI documentation.

ACADEMIC PROJECT

Substrate Analysis on the Design of Wide-band Antenna for 5G Sub-6 GHz Communication | 2024–2026

- Used CST to simulate and iteratively optimize substrate design for bandwidth, gain and return loss, analyzing S11, VSWR and efficiency.

TRAINING & CERTIFICATION

Railway Training under Sealdah Division (07 July 2025 to 31 July 2025)

- Trained in Electronic Interlocking (EI), Telecom works (OFC), LC Gate, UTS and PRS at Eastern Railway, Sealdah; certified by Sr. DSTE.

INTERESTS

Cricket | Football | Recitation